

A value-iteration proof of threshold optimality for one fast server and two identical slow servers

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Abstract

We prove a threshold theorem for the Poisson-arrival model with one fast exponential server and two identical slower exponential servers. The state is (x, y) , where x is the number of customers in the queue plus the customer, if any, at the fast server, and $y \in \{0, 1, 2\}$ is the number of busy slow servers. The service rates are $\mu_1 > \mu_2 > 0$, the arrival rate is λ , and the one-step cost is the number of customers in the system. The proof uses uniformization, a relaxed tail-minimum assignment operator, and an invariant cone of lattice inequalities. The additional inequality, relative to the classical two-server proof, is convexity in the number of busy slow servers at fixed total population. A boundary-dominance argument then shows that the relaxed operator never needs to move a fast-server customer to a slow server, so the relaxed and true nonpreemptive recursions coincide on the value iterates. The proof also isolates why the true masked nonpreemptive assignment operator is not cone-closed on arbitrary cone functions. The conclusion is that, for each slow-server occupancy $y = 0, 1$, the set of queue lengths at which it is optimal to send a queued customer to an idle slow server is an upper set. When both slow servers are idle, an optimal selector has ordered thresholds: use no slow server, then one slow server, then two slow servers.

1 Model and statement

There is one fast server of rate μ_1 and two identical slow servers of rate μ_2 , with

$$\mu_1 > \mu_2 > 0. \quad (1.1)$$

Arrivals are Poisson with rate λ . Customers are assigned nonpreemptively. Once a customer is assigned to a server, that customer cannot be moved to another server. The holding cost rate is the number of customers in the system.

The full server-state description is reduced in the usual way: whenever the fast server is idle and a customer is available for assignment, there is an optimal policy that assigns a customer to the fast server before assigning any customer to a slow server. Thus it is enough to use the state

$$(x, y) \in \mathbb{Z}_+ \times \{0, 1, 2\}, \quad (1.2)$$

where x is the number of customers in the queue plus the customer at the fast server, and y is the number of busy slow servers. The total number of customers is $x + y$, and the one-step holding cost is

$$c(x, y) = x + y. \quad (1.3)$$

The number of actually waiting customers is

$$q(x) = (x - 1)^+. \quad (1.4)$$

Consequently, an assignment to a slow server may only use one of these $q(x)$ queued customers.

Lemma 1 (Fast-server reduction). *For the original nonpreemptive system, there is an optimal policy that never routes a customer to an idle slow server while the fast server is idle and a customer is available for service.*

Proof. It is enough to show that any policy that violates the stated rule can be weakly improved by an admissible history-dependent policy. Call the system governed by the original policy π the π -system, and call the system governed by the modified policy $\tilde{\pi}$ the $\tilde{\pi}$ -system. Let $X^\pi(t)$ and $\tilde{X}(t)$ denote their real-customer state processes, and let $|X^\pi(t)|$ and $|\tilde{X}(t)|$ denote the corresponding numbers of real customers in system.

Consider the first decision epoch at which π sends a queued customer C to an idle slow server S while the fast server F is idle. Shift time so that this epoch is time 0. The policy $\tilde{\pi}$ sends C to F at time 0. Apart from this interchange, $\tilde{\pi}$ follows the decisions prescribed by π on an internally simulated copy of the π -system, translating them into feasible actions for the $\tilde{\pi}$ -system as described below. Arrivals, services outside the interchange, and any randomization used by π are coupled identically whenever the two systems have been synchronized.

Couple the possible service times of the tagged customer C by taking

$$E \sim \text{Exp}(1) \tag{1.5}$$

and setting

$$\tau_F = \frac{E}{\mu_1}, \quad \tau_S = \frac{E}{\mu_2}. \tag{1.6}$$

Then

$$\tau_F \sim \text{Exp}(\mu_1), \quad \tau_S \sim \text{Exp}(\mu_2), \quad \tau_F \leq \tau_S \quad \text{a.s.} \tag{1.7}$$

Let T be the first time before τ_S at which the π -system would assign a real customer to the fast server F ; if this never occurs before τ_S , set $T = \infty$.

Case 1: the modified fast service completes first. Assume

$$\tau_F \leq T. \tag{1.8}$$

Then C completes service in the $\tilde{\pi}$ -system before the π -system uses F . At time τ_F , the $\tilde{\pi}$ -system has one fewer real customer than the π -system. The auxiliary comparison now reserves the slow server S in the $\tilde{\pi}$ -system during

$$[\tau_F, \tau_S), \tag{1.9}$$

meaning that $\tilde{\pi}$ treats S as unavailable during this interval. This is the same as placing a nonpaying dummy customer on S for comparison purposes. The dummy is not a real customer and is not counted in the holding cost. Since idling a server is an admissible action, this reservation can be implemented by the actual policy $\tilde{\pi}$.

During $[\tau_F, \tau_S)$, the π -system still has the real customer C on S , while the $\tilde{\pi}$ -system has only the dummy reservation on S . Every action of π that uses F can be copied by $\tilde{\pi}$, because F is available in the $\tilde{\pi}$ -system whenever it is available in the π -system. At time τ_S , customer C completes in the π -system and the reservation is removed in the $\tilde{\pi}$ -system. From that time on, the two real-customer systems can be run identically.

Case 2: the original policy activates the fast server before C completes under $\tilde{\pi}$. Assume

$$T < \tau_F. \tag{1.10}$$

At time T , the π -system has C on S and assigns some real customer D to F . In the $\tilde{\pi}$ -system, C is still on F and S is idle, so $\tilde{\pi}$ assigns the corresponding customer D to S . At time T , the two

systems have the same number of real customers and the same busy/idle server pattern; only the labels C and D have been interchanged between F and S .

By the memoryless property, at the stopping time T the unused service times on the busy fast servers may be restarted with a common exponential residual of rate μ_1 , and the unused service times on the busy slow servers may be restarted with a common exponential residual of rate μ_2 . Thus, after relabeling the customers, the two systems can be coupled by physical server identity from time T onward. Labels do not affect holding costs or feasible actions, so their future real-customer counts and future costs can be made equal.

In both cases, for every sample path and every time t ,

$$|\tilde{X}(t)| \leq |X^\pi(t)|, \quad (1.11)$$

where dummy reservations are not counted. Hence the accumulated holding cost of $\tilde{\pi}$ on every finite time interval, and therefore also its discounted or long-run average holding cost whenever these criteria are defined, is no larger than that of π .

The modified policy is legitimate: it is a history-dependent policy with finite internal memory that simulates the π -system, records whether the interchange is active, and, in Case 1, reserves S until the privately computed time $\tau_S = (\mu_1/\mu_2)\tau_F$ after observing the completion time τ_F . Its decisions use only the observed history, the rule π , and private randomization; they do not depend on future arrivals or future real service completions. Repeating this interchange whenever a violation would occur gives a policy with no larger cost that always uses the fast server before assigning a customer to any idle slow server. Thus some optimal policy has the stated fast-first property. $\square$

Uniformized dynamic program

We use uniformization. A harmless fictitious self-transition can be added so that the total transition probability is one. Self-transition terms are multiples of the identity operator and preserve every inequality used below; they are suppressed in the displayed formulae. Equivalently, one may read the coefficients λ, μ_1, μ_2 below as nonnegative uniformized transition weights after division by a common constant. For discounted costs an additional discount factor $\rho \in (0, 1)$ multiplies the continuation operator; for finite-horizon undiscounted value iteration take $\rho = 1$.

Let V_m be the m -step value function after decisions have been made. Write the recursion as

$$V_{m+1} = c + \rho \mathcal{P}QV_m. \quad (1.12)$$

The event operator $\mathcal{P}$ is

$$(\mathcal{P}f)(x, 0) = \lambda f(x+1, 0) + \mu_1 f((x-1)^+, 0) + 2\mu_2 f(x, 0), \quad (1.13)$$

$$(\mathcal{P}f)(x, 1) = \lambda f(x+1, 1) + \mu_1 f((x-1)^+, 1) + \mu_2 f(x, 0) + \mu_2 f(x, 1), \quad (1.14)$$

$$(\mathcal{P}f)(x, 2) = \lambda f(x+1, 2) + \mu_1 f((x-1)^+, 2) + 2\mu_2 f(x, 1). \quad (1.15)$$

Here $(x-1)^+$ is the reflected fast-server-completion boundary.

The true nonpreemptive assignment operator is

$$(\mathcal{Q}^{\text{true}}f)(x, y) = \min_{0 \leq a \leq \min\{2-y, (x-1)^+\}} f(x-a, y+a). \quad (1.16)$$

It assigns only customers who are actually waiting in queue. In particular,

$$(\mathcal{Q}^{\text{true}}f)(1, 0) = f(1, 0), \quad (\mathcal{Q}^{\text{true}}f)(1, 1) = f(1, 1), \quad (1.17)$$

and

$$(\mathcal{Q}^{\text{true}}f)(2, 0) = \min\{f(2, 0), f(1, 1)\}. \quad (1.18)$$

The state $f(0, 1)$ is meaningful; it is not made infinite. What is infeasible is only the assignment arc $(1, 0) \rightarrow (0, 1)$.

The proof first works with the relaxed tail-minimum operator

$$(\mathcal{Q}f)(x, y) = \min_{y \leq k \leq \min\{2, x+y\}} f(x + y - k, k). \quad (1.19)$$

This operator minimizes over all states with the same total population and no fewer busy slow servers. It differs from the true nonpreemptive operator only at the small-population boundary, where it formally permits moving a fast-server customer to a slow server. Section 5 proves that these relaxed boundary actions are never used by the value functions generated by (1.12). Hence the relaxed and true recursions coincide on the value iterates.

Theorem 1 (Threshold optimality). *For the Poisson-arrival one-fast/two-identical-slow system described above, there is an optimal stationary discounted policy with the following structure.*

- (i) *If one slow server is busy and the other is idle, the set of x for which it is optimal to assign a queued customer to the idle slow server is an upper set.*
- (ii) *If both slow servers are idle, an optimal selector has ordered thresholds: as x increases, the selector uses no slow server, then exactly one slow server, and then both slow servers, with the obvious infeasible small- x actions suppressed.*

If $\lambda < \mu_1 + 2\mu_2$, an average-cost optimal stationary policy with the same threshold structure exists.

2 The invariant inequalities

It is convenient to use total-population coordinates. For a function f on states, define

$$H_y(n) = f(n - y, y), \quad y = 0, 1, 2, \quad (2.1)$$

for $n \geq y$. Thus $n = x + y$ is the total number of customers and the subscript y is the number of busy slow servers. Put

$$d_y(n) = H_y(n + 1) - H_y(n), \quad n \geq y. \quad (2.2)$$

We say that $H \in \mathcal{K}$ if the following seven inequalities hold on the domains displayed below. For (K1)–(K4), both the H -form and the d -form are shown, because the H -form is useful for boundary checks whereas the d -form is useful for slope comparisons.

(K1) Monotonicity in n fixing y

$$\begin{aligned} H_y(n+1) - H_y(n) &\geq 0 \\ \Leftrightarrow d_y(n) &\geq 0, \end{aligned} \quad y = 0, 1, 2, n \geq y.$$

(K2) Convexity in n fixing y

$$\begin{aligned} H_y(n+2) + H_y(n) - 2H_y(n+1) &\geq 0 \\ \Leftrightarrow d_y(n+1) - d_y(n) &\geq 0, \end{aligned} \quad y = 0, 1, 2, n \geq y.$$

(K3) Left multimodularity

$$\begin{aligned} H_y(n+1) - H_y(n) - H_{y+1}(n+1) + H_{y+1}(n) &\geq 0 \\ \Leftrightarrow d_y(n) - d_{y+1}(n) &\geq 0, \end{aligned} \quad y = 0, 1, n \geq y+1.$$

(K4) Right multimodularity

$$\begin{aligned} H_{y+1}(n+2) - H_{y+1}(n+1) - H_y(n+1) + H_y(n) &\geq 0 \\ \Leftrightarrow d_{y+1}(n+1) - d_y(n) &\geq 0, \end{aligned} \quad y = 0, 1, n \geq y.$$

(K5) Monotonicity in y fixing x

$$H_{y+1}(n+1) - H_y(n) \geq 0, \quad y = 0, 1, n \geq y.$$

(K6) Convexity in y fixing x

$$H_0(n) + H_2(n+2) - 2H_1(n+1) \geq 0, \quad n \geq 0.$$

(K7) Convexity in y fixing n

$$H_0(n) + H_2(n) - 2H_1(n) \geq 0. \quad n \geq 2.$$

Inequalities (K1)–(K2) are monotonicity and convexity in total population along each slow-server level. Inequalities (K3)–(K4) are the two adjacent multimodularity inequalities. Inequality (K5) says that replacing a queue/fast-coordinate customer by a slow-service customer, while increasing total population by one, does not decrease the value. Inequality (K6) is convexity in the number of slow busy servers at fixed fast-coordinate x . Inequality (K7) is convexity in the number of slow busy servers at fixed total population; this is the additional inequality used here.

The cost $c(x, y) = x + y$ lies in $\mathcal{K}$, since in total-population coordinates $H_y(n) = n$ for all y .

For later use define the fixed-total differences

$$A(n) = H_0(n) - H_1(n), \quad n \geq 1, \quad (2.3)$$

and

$$B(n) = H_1(n) - H_2(n), \quad n \geq 2. \quad (2.4)$$

Then (K3) says that A and B are nondecreasing, because

$$A(n+1) - A(n) = d_0(n) - d_1(n), \quad (2.5)$$

for $n \geq 1$, and

$$B(n+1) - B(n) = d_1(n) - d_2(n), \quad (2.6)$$

for $n \geq 2$. Inequality (K7) says

$$A(n) \geq B(n), \quad n \geq 2. \quad (2.7)$$

We record three elementary closure properties of $\mathcal{K}$ that will be used in the discounted and average-cost passage. First, if $H, G \in \mathcal{K}$ and $a, b \geq 0$, then $aH + bG \in \mathcal{K}$. Second, if $H \in \mathcal{K}$ and

$r \in \mathbb{R}$, then $H + r \in \mathcal{K}$, where $(H + r)_y(n) = H_y(n) + r$ for all admissible (y, n) . Third, if $H^m \in \mathcal{K}$ for all m and $H_y^m(n) \rightarrow H_y(n)$ for every admissible (y, n) , then $H \in \mathcal{K}$. Indeed, each defining inequality (K1)–(K7) is a homogeneous linear inequality in finitely many coordinates of H , which gives closure under nonnegative linear combinations. In each inequality the coefficients sum to zero, so adding a common constant to all coordinates leaves the left-hand side unchanged. Finally, pointwise limits preserve the inequalities because the limit of nonnegative real numbers is nonnegative. Consequently, positive rescaling and subtracting a constant preserve membership in $\mathcal{K}$. The boundary comparisons $V(1, 0) \leq V(0, 1)$ and $V(1, 1) \leq V(0, 2)$ are also preserved by positive rescaling and by subtracting the same constant from all state values.

3 Closure under assignment

In total-population coordinates the relaxed assignment operator (1.19) is the tail minimum

$$(\mathcal{Q}H)_y(n) = \min_{y \leq k \leq \min\{2, n\}} H_k(n). \quad (3.1)$$

Warning 1 (The true masked operator is not cone-closed). *The true nonpreemptive operator $\mathcal{Q}^{\text{true}}$ does not map $\mathcal{K}$ into $\mathcal{K}$ on arbitrary functions. Take*

$$f(x, y) = x. \quad (3.2)$$

Then $H_y(n) = n - y$, and all inequalities (K1)–(K7) hold. However,

$$(\mathcal{Q}^{\text{true}}f)(0, 0) = 0, \quad (3.3)$$

$$(\mathcal{Q}^{\text{true}}f)(1, 0) = 1, \quad (3.4)$$

and

$$(\mathcal{Q}^{\text{true}}f)(2, 0) = 1, \quad (3.5)$$

because from $(2, 0)$ one queued customer can be assigned to one slow server. Therefore

$$(\mathcal{Q}^{\text{true}}f)(2, 0) - 2(\mathcal{Q}^{\text{true}}f)(1, 0) + (\mathcal{Q}^{\text{true}}f)(0, 0) = -1, \quad (3.6)$$

so convexity in x fails. Thus the relaxed tail-minimum operator is not merely a notational convenience; the proof must use the relaxed operator for cone closure and then prove separately that infeasible relaxed boundary arcs are not selected by the value iterates.

Lemma 2 (Assignment closure). *If $H \in \mathcal{K}$, then $\mathcal{Q}H \in \mathcal{K}$.*

Proof. Write $\bar{H} = \mathcal{Q}H$. First define the selectors by the tail-minimum itself. For each admissible pair (n, y) , let

$$r_y(n) = \min \arg \min_{y \leq k \leq \min\{2, n\}} H_k(n), \quad (3.7)$$

where the minimum outside the $\arg \min$ is the deterministic smallest-index tie convention. Then

$$\bar{H}_y(n) = H_{r_y(n)}(n). \quad (3.8)$$

The sign formulae below are only shorthand for the interior cases in which the corresponding differences are defined.

For $y = 0$, the boundary values are

$$r_0(0) = 0, \quad (3.9)$$

and

$$r_0(1) = \begin{cases} 0, & A(1) \leq 0, \\ 1, & A(1) > 0. \end{cases} \quad (3.10)$$

For $n \geq 2$,

$$r_0(n) = \begin{cases} 0, & A(n) \leq 0, \\ 1, & A(n) > 0 \text{ and } B(n) \leq 0, \\ 2, & B(n) > 0. \end{cases} \quad (3.11)$$

For $y = 1$,

$$r_1(1) = 1, \quad (3.12)$$

and, for $n \geq 2$,

$$r_1(n) = \begin{cases} 1, & B(n) \leq 0, \\ 2, & B(n) > 0. \end{cases} \quad (3.13)$$

Finally,

$$r_2(n) = 2, \quad n \geq 2. \quad (3.14)$$

Because A and B are nondecreasing and $A(n) \geq B(n)$ for $n \geq 2$, the selectors r_0, r_1, r_2 are nondecreasing in n . The only boundary points that require comment are $r_0(0) \leq r_0(1)$, which is immediate, and $r_0(1) \leq r_0(2)$. If $r_0(1) = 0$, this is immediate; if $r_0(1) = 1$, then $A(1) > 0$, so $A(2) > 0$ by monotonicity of A , and $r_0(2)$ cannot be 0. Also $r_1(1) = 1 \leq r_1(2)$.

(K3) and (K7). The desired (K3) inequalities for $\bar{H}$ are

$$[\bar{H}_y(n+1) - \bar{H}_y(n)] - [\bar{H}_{y+1}(n+1) - \bar{H}_{y+1}(n)] \geq 0, \quad (3.15)$$

for $y = 0, 1$ on their natural domains. Equivalently, for $y = 0$ we need $\bar{A}(n+1) - \bar{A}(n) \geq 0$, and for $y = 1$ we need $\bar{B}(n+1) - \bar{B}(n) \geq 0$, where

$$\bar{A}(n) := \bar{H}_0(n) - \bar{H}_1(n), \quad n \geq 1, \quad (3.16)$$

and

$$\bar{B}(n) := \bar{H}_1(n) - \bar{H}_2(n), \quad n \geq 2. \quad (3.17)$$

From the tail-minimum definition, and because (K7) holds for H ,

$$\bar{A}(n) = \min\{A(n), 0\}, \quad n \geq 1. \quad (3.18)$$

Similarly, from the tail-minimum definition,

$$\bar{B}(n) = \min\{B(n), 0\}, \quad n \geq 2. \quad (3.19)$$

Since A and B are nondecreasing, so are $\bar{A}$ and $\bar{B}$, proving (K3).

The desired (K7) inequality for $\bar{H}$ is

$$\bar{H}_0(n) + \bar{H}_2(n) - 2\bar{H}_1(n) \geq 0, \quad n \geq 2, \quad (3.20)$$

which is equivalent to $\bar{A}(n) \geq \bar{B}(n)$. If $A(n) \leq 0$, then $\bar{A}(n) = A(n) \geq B(n) = \bar{B}(n)$; if $A(n) > 0$ and $B(n) \leq 0$, then $\bar{A}(n) = 0 \geq B(n) = \bar{B}(n)$; if $B(n) > 0$, then both $\bar{A}(n)$ and $\bar{B}(n)$ are zero. Thus (K7) holds.

Small- n boundary cases. There is no $\bar{B}(1)$, because $\bar{H}_2(1)$ is not defined. This causes no gap: (K3) with $y = 0$ begins at $n = 1$ and uses only $\bar{A}(1), \bar{A}(2), \dots$; (K3) with $y = 1$ begins at $n = 2$ and uses only $\bar{B}(2), \bar{B}(3), \dots$; and (K7) begins at $n = 2$. Thus no condition in the cone requires $B(1)$ or $\bar{B}(1)$.

It remains to verify (K1), (K2), (K4), (K5), and (K6). The verification is finite because selected indices can only be 0, 1, 2 and the selectors are nondecreasing. In the tables below, the phrase “selected i at m ” means $H_i(m) \leq H_k(m)$ for every feasible k in the relevant tail. All inequalities are weak, so ties cause no difficulty.

Monotonicity in n (K1). The desired (K1) inequality for $\bar{H}$ is

$$\bar{H}_y(n+1) - \bar{H}_y(n) \geq 0, \quad y = 0, 1, 2. \quad (3.21)$$

Let

$$r = r_y(n), \quad s = r_y(n+1). \quad (3.22)$$

Using (3.8), (3.21) becomes

$$H_s(n+1) - H_r(n) \geq 0. \quad (3.23)$$

Because each selector r_y is nondecreasing, every actual pair satisfies $r \leq s$. Conversely, every instance of (K1) for $\bar{H}$ is obtained in this way from the adjacent selector pair $(r_y(n), r_y(n+1))$. Hence it suffices to check (3.23) for the possible nondecreasing adjacent selector pairs.

Small- n boundary cases. For $y = 0, n = 0$, the first tail contains only index 0, so the only possible pairs are (0, 0) and (0, 1). For $y = 0, n = 1$, the possible pairs are a subset of (0, 0), (0, 1), (0, 2), (1, 1), (1, 2). For $y = 0, n \geq 2$, all six pairs listed for $y = 0$ in the table can occur. For $y = 1, n = 1$, the only possible pairs are (1, 1) and (1, 2); for $y = 1, n \geq 2$, the possible pairs are (1, 1), (1, 2), (2, 2). For $y = 2$, only (2, 2) occurs. Thus the table checks a superset of the boundary-feasible cases, and every term in an actually occurring row is defined.

y	$(r_y(n), r_y(n+1))$	Desired term $H_s(n+1) - H_r(n)$	Reason
0	(0, 0)	$H_0(n+1) - H_0(n)$	(K1) for H_0
0	(0, 1)	$H_1(n+1) - H_0(n)$	(K5), $y = 0$
0	(0, 2)	$H_2(n+1) - H_0(n)$	$[H_2(n+1) - H_1(n)] + [H_1(n) - H_0(n)] \geq 0$
0	(1, 1)	$H_1(n+1) - H_1(n)$	(K1) for H_1
0	(1, 2)	$H_2(n+1) - H_1(n)$	(K5), $y = 1$
0	(2, 2)	$H_2(n+1) - H_2(n)$	(K1) for H_2
1	(1, 1)	$H_1(n+1) - H_1(n)$	already checked
1	(1, 2)	$H_2(n+1) - H_1(n)$	already checked
1	(2, 2)	$H_2(n+1) - H_2(n)$	already checked
2	(2, 2)	$H_2(n+1) - H_2(n)$	already checked

In the row (0, 2), the first bracket is nonnegative by (K5) with $y = 1$, and the second is nonnegative because selecting index 0 at total population n gives $H_0(n) \leq H_1(n)$.

Convexity in n (K2). The desired (K2) inequality for $\bar{H}$ is

$$[\bar{H}_y(n+2) - \bar{H}_y(n+1)] - [\bar{H}_y(n+1) - \bar{H}_y(n)] \geq 0, \quad (3.24)$$

or equivalently

$$\bar{H}_y(n) + \bar{H}_y(n+2) - 2\bar{H}_y(n+1) \geq 0. \quad (3.25)$$

Let

$$r = r_y(n), \quad s = r_y(n+1), \quad t = r_y(n+2). \quad (3.26)$$

Using (3.8), (3.25) becomes

$$H_r(n) + H_t(n+2) - 2H_s(n+1) \geq 0. \quad (3.27)$$

Because r_y is nondecreasing, every actual selector triple satisfies $r \leq s \leq t$. Conversely, every instance of (K2) for $\bar{H}$ arises from such a consecutive selector triple. Hence it suffices to verify (3.27) for the feasible nondecreasing triples.

Small- n boundary cases. For $y = 0, n = 0$, the first tail contains only 0, the second tail contains 0, 1, and the third tail contains 0, 1, 2; hence only the $y = 0$ rows with first coordinate 0 can occur, and (0, 2, 2) is impossible because index 2 is not feasible at total population 1. For $y = 0, n = 1$, the first tail contains 0, 1 and the later tails contain all three indices, so no triple with first coordinate 2 can occur. For $y = 0, n \geq 2$, all $y = 0$ rows in the table may occur. For $y = 1, n = 1$, only (1, 1, 1), (1, 1, 2), (1, 2, 2) can occur; for $y = 1, n \geq 2$, also (2, 2, 2) can occur. For $y = 2$, only (2, 2, 2) occurs. Thus impossible boundary triples are not used, and all terms in each actually occurring row are defined.

y	$(r_y(n),$ $r_y(n+1),$ $r_y(n+2))$	Left side of (3.27)	Reason
0	(0, 0, 0)	$H_0(n) + H_0(n+2) - 2H_0(n+1)$	(K2) for H_0
0	(0, 0, 1)	$H_0(n) + H_1(n+2) - 2H_0(n+1)$	$H_1(n+2) - H_0(n+1) \geq d_1(n+1) \geq d_0(n)$
0	(0, 0, 2)	$H_0(n) + H_2(n+2) - 2H_0(n+1)$	(K6) $+2[H_1(n+1) - H_0(n+1)] \geq 0$
0	(0, 1, 1)	$H_0(n) + H_1(n+2) - 2H_1(n+1)$	$H_1(n+1) - H_0(n) \leq d_0(n) \leq d_1(n+1)$
0	(0, 1, 2)	$H_0(n) + H_2(n+2) - 2H_1(n+1)$	(K6)
0	(0, 2, 2)	$H_0(n) + H_2(n+2) - 2H_2(n+1)$	(K6) $+2[H_1(n+1) - H_2(n+1)] \geq 0$
0	(1, 1, 1)	$H_1(n) + H_1(n+2) - 2H_1(n+1)$	(K2) for H_1
0	(1, 1, 2)	$H_1(n) + H_2(n+2) - 2H_1(n+1)$	$H_2(n+2) - H_1(n+1) \geq d_2(n+1) \geq d_1(n)$
0	(1, 2, 2)	$H_1(n) + H_2(n+2) - 2H_2(n+1)$	$H_2(n+1) - H_1(n) \leq d_1(n) \leq d_2(n+1)$
0	(2, 2, 2)	$H_2(n) + H_2(n+2) - 2H_2(n+1)$	(K2) for H_2
1	(1, 1, 1)	$H_1(n) + H_1(n+2) - 2H_1(n+1)$	already checked
1	(1, 1, 2)	$H_1(n) + H_2(n+2) - 2H_1(n+1)$	already checked
1	(1, 2, 2)	$H_1(n) + H_2(n+2) - 2H_2(n+1)$	already checked
1	(2, 2, 2)	$H_2(n) + H_2(n+2) - 2H_2(n+1)$	already checked
2	(2, 2, 2)	$H_2(n) + H_2(n+2) - 2H_2(n+1)$	already checked

The inequalities $d_1(n+1) \geq d_0(n)$ and $d_2(n+1) \geq d_1(n)$ are (K4). The extra nonnegative terms involving selected values follow from the definition of the selected index at the relevant total population.

Right multimodularity (K4). The desired (K4) inequality for $\bar{H}$ is

$$\bar{H}_y(n) + \bar{H}_{y+1}(n+2) - \bar{H}_y(n+1) - \bar{H}_{y+1}(n+1) \geq 0, \quad (3.28)$$

for $y = 0, 1$ on the natural domains.

First take $y = 1$. Since $\bar{H}_2 = H_2$, (3.28) is

$$\bar{H}_1(n) + H_2(n+2) - \bar{H}_1(n+1) - H_2(n+1) \geq 0, \quad n \geq 1, \quad (3.29)$$

equivalently,

$$d_2(n+1) \geq \bar{H}_1(n+1) - \bar{H}_1(n). \quad (3.30)$$

Small- n boundary cases for $y = 1$. At $n = 1$, $B(1)$ is not defined and $\bar{H}_1(1) = H_1(1)$. If $\bar{H}_1(2) = H_1(2)$, then (3.30) is $d_2(2) \geq d_1(1)$, which is (K4) for H with $y = 1, n = 1$. If $\bar{H}_1(2) = H_2(2)$, then $H_2(2) \leq H_1(2)$, so

$$\bar{H}_1(2) - \bar{H}_1(1) = H_2(2) - H_1(1) \leq H_1(2) - H_1(1) = d_1(1), \quad (3.31)$$

and (K4) again gives $d_2(2) \geq d_1(1)$.

For $y = 1$ and $n \geq 2$, $B(n)$ and $B(n+1)$ are defined. If $B(n) \leq 0$ and $B(n+1) \leq 0$, then the right side of (3.30) is $d_1(n)$, and (K4) for H with $y = 1$ applies. If $B(n) \leq 0 < B(n+1)$, the right side is $H_2(n+1) - H_1(n) \leq d_1(n)$ because $H_2(n+1) \leq H_1(n+1)$; again (K4) gives the result. If $B(n) > 0$, the right side is $d_2(n)$, and convexity (K2) for H_2 gives $d_2(n+1) \geq d_2(n)$.

Now take $y = 0$. The desired inequality (3.28) is

$$\bar{H}_0(n) + \bar{H}_1(n+2) - \bar{H}_0(n+1) - \bar{H}_1(n+1) \geq 0, \quad n \geq 0, \quad (3.32)$$

equivalently,

$$\bar{H}_1(n+2) - \bar{H}_1(n+1) \geq \bar{H}_0(n+1) - \bar{H}_0(n). \quad (3.33)$$

Small- n boundary cases for $y = 0$. At $n = 0$, $B(n+1)$ is not defined, $\bar{H}_0(0) = H_0(0)$, and $\bar{H}_1(1) = H_1(1)$. If $\bar{H}_0(1) = H_0(1)$ and $\bar{H}_1(2) = H_1(2)$, then (3.33) follows from $d_1(1) \geq d_0(0)$, which is (K4) for H with $y = 0, n = 0$. If $\bar{H}_0(1) = H_1(1)$ and $\bar{H}_1(2) = H_1(2)$, then

$$\bar{H}_0(1) - \bar{H}_0(0) = H_1(1) - H_0(0) \leq H_0(1) - H_0(0) = d_0(0), \quad (3.34)$$

and again (K4) gives $d_1(1) \geq d_0(0)$. If $\bar{H}_1(2) = H_2(2)$, then (K6) at $n = 0$ gives

$$H_2(2) - H_1(1) \geq H_1(1) - H_0(0). \quad (3.35)$$

If $\bar{H}_0(1) = H_1(1)$, this is exactly (3.33). If $\bar{H}_0(1) = H_0(1)$, then $H_0(1) \leq H_1(1)$, so

$$H_2(2) - H_1(1) \geq H_1(1) - H_0(0) \geq H_0(1) - H_0(0), \quad (3.36)$$

which again proves (3.33).

For $y = 0$ and $n \geq 1$, the needed $B(n+1)$ and $B(n+2)$ are defined. If $B(n+1) \geq 0$, then as values $\bar{H}_1(n+1) = H_2(n+1)$ and $\bar{H}_1(n+2) = H_2(n+2)$. Also $\bar{H}_0(n+1) = H_2(n+1)$. If $\bar{H}_0(n) = H_2(n)$, (3.33) follows from (K2) for H_2 . If $\bar{H}_0(n) = H_1(n)$, it follows from

$$H_2(n+1) - H_1(n) \leq d_1(n) \leq d_2(n+1), \quad (3.37)$$

where the last inequality is (K4) with $y = 1$. If $\bar{H}_0(n) = H_0(n)$, then (K6) and $H_2(n+1) \leq H_1(n+1)$ give

$$H_0(n) + H_2(n+2) - 2H_2(n+1) \geq 0, \quad (3.38)$$

which is exactly $d_2(n+1) \geq H_2(n+1) - H_0(n)$.

If $B(n+1) < 0$ and $B(n+2) < 0$, then the left side of (3.33) is $d_1(n+1)$ and $\bar{H}_0(n), \bar{H}_0(n+1)$ can only select 0 or 1. The three nondecreasing possibilities are (0, 0), (0, 1), and (1, 1). For (0, 0), (K4) with $y = 0$ gives $d_1(n+1) \geq d_0(n)$. For (0, 1),

$$H_1(n+1) - H_0(n) \leq H_0(n+1) - H_0(n) = d_0(n), \quad (3.39)$$

because selecting 1 at $n+1$ gives $H_1(n+1) \leq H_0(n+1)$; then use (K4) with $y=0$. For $(1,1)$, (K2) for H_1 gives $d_1(n+1) \geq d_1(n)$.

Finally suppose $B(n+1) < 0 \leq B(n+2)$. Then the left side of (3.33) is $H_2(n+2) - H_1(n+1)$. Again $\bar{H}_0(n), \bar{H}_0(n+1)$ can only select 0 or 1. For $(0,0)$, selection of 0 at $n+1$ gives

$$H_1(n+1) - H_0(n) \geq H_0(n+1) - H_0(n) = d_0(n), \quad (3.40)$$

and (K6) gives

$$H_2(n+2) - H_1(n+1) \geq H_1(n+1) - H_0(n) \geq d_0(n). \quad (3.41)$$

For $(0,1)$, (K6) directly gives

$$H_2(n+2) - H_1(n+1) \geq H_1(n+1) - H_0(n). \quad (3.42)$$

For $(1,1)$,

$$H_2(n+2) - H_1(n+1) = d_2(n+1) - B(n+1) \geq d_2(n+1) \geq d_1(n), \quad (3.43)$$

using $B(n+1) < 0$ and (K4) with $y=1$. This proves (K4) for $\bar{H}$.

Monotonicity in y fixing x (K5). The desired (K5) inequality for $\bar{H}$ is

$$\bar{H}_{y+1}(n+1) - \bar{H}_y(n) \geq 0, \quad y = 0, 1. \quad (3.44)$$

For $y=0$, this compares the selected values $r_0(n)$ and $r_1(n+1)$; for $y=1$, it compares $r_1(n)$ and $r_2(n+1) = 2$.

Small- n boundary cases. For $y=0, n=0$, the only possible pair is $(r_0(0), r_1(1)) = (0, 1)$. For $y=0, n=1$, the possible pairs are $(0, 1), (0, 2), (1, 1), (1, 2)$; for $y=0, n \geq 2$, the pair $(2, 2)$ can also occur. For $y=1, n=1$, the only possible pair is $(1, 2)$; for $y=1, n \geq 2$, the possible pairs are $(1, 2)$ and $(2, 2)$.

y	$(r_y(n), r_{y+1}(n+1))$	Desired term $\bar{H}_{y+1}(n+1) - \bar{H}_y(n)$	Reason
0	(0, 1)	$H_1(n+1) - H_0(n)$	(K5), $y=0$
0	(0, 2)	$H_2(n+1) - H_0(n)$	$[H_2(n+1) - H_1(n)] + [H_1(n) - H_0(n)] \geq 0$
0	(1, 1)	$H_1(n+1) - H_1(n)$	(K1) for H_1
0	(1, 2)	$H_2(n+1) - H_1(n)$	(K5), $y=1$
0	(2, 2)	$H_2(n+1) - H_2(n)$	(K1) for H_2
1	(1, 2)	$H_2(n+1) - H_1(n)$	already checked
1	(2, 2)	$H_2(n+1) - H_2(n)$	already checked

The pair $(2, 1)$ in the $y=0$ comparison is impossible because B is nondecreasing; as a value statement, if equality occurs at a tie then either representative gives the same $\bar{H}$.

Convexity in y fixing x (K6). The desired (K6) inequality for $\bar{H}$ is

$$\bar{H}_0(n) + \bar{H}_2(n+2) - 2\bar{H}_1(n+1) \geq 0, \quad n \geq 0. \quad (3.45)$$

Since $\bar{H}_2 = H_2$, this is

$$\bar{H}_0(n) + H_2(n+2) - 2\bar{H}_1(n+1) \geq 0. \quad (3.46)$$

Small- n boundary cases. For $n = 0$, the only possible pair $(r_0(0), r_1(1))$ is $(0, 1)$, so (3.46) is exactly (K6) for H at $n = 0$. For $n = 1$, the possible pairs are $(0, 1), (0, 2), (1, 1), (1, 2)$; the pair $(2, 2)$ cannot occur because $H_2(1)$ is not defined. For $n \geq 2$, all five pairs in the table may occur.

$(r_0(n), r_1(n+1))$	Left side of (3.46)	Reason
$(0, 1)$	$H_0(n) + H_2(n+2) - 2H_1(n+1)$	(K6)
$(0, 2)$	$H_0(n) + H_2(n+2) - 2H_2(n+1)$	(K6) + $2[H_1(n+1) - H_2(n+1)] \geq 0$
$(1, 1)$	$H_1(n) + H_2(n+2) - 2H_1(n+1)$	$H_2(n+2) - H_1(n+1) \geq d_2(n+1) \geq d_1(n)$
$(1, 2)$	$H_1(n) + H_2(n+2) - 2H_2(n+1)$	$H_2(n+1) - H_1(n) \leq d_1(n) \leq d_2(n+1)$
$(2, 2)$	$H_2(n) + H_2(n+2) - 2H_2(n+1)$	(K2) for H_2

This proves all defining inequalities for $\bar{H}$, so $\mathcal{QH} \in \mathcal{K}$. □

4 Closure under event transitions

Lemma 3 (Event closure). *If $f \in \mathcal{K}$, then $\mathcal{P}f \in \mathcal{K}$.*

Proof. Let $g = \mathcal{P}f$. We verify the event operator in the original (x, y) coordinates. For any function h on the state space, write

$$h_y(x) = h(x, y), \quad y = 0, 1, 2. \quad (4.1)$$

Define the primary finite-difference quantities of h , on their natural domains, by

$$X_y^h(x) = h_y(x+1) - h_y(x), \quad y = 0, 1, 2, \quad x \geq 0, \quad (4.2)$$

$$C_y^h(x) = h_y(x+1) - 2h_y(x) + h_y(x-1), \quad y = 0, 1, 2, \quad x \geq 1, \quad (4.3)$$

$$Y_y^h(x) = h_{y+1}(x) - h_y(x), \quad y = 0, 1, \quad x \geq 0, \quad (4.4)$$

$$Z^h(x) = h_0(x) + h_2(x) - 2h_1(x), \quad x \geq 0, \quad (4.5)$$

$$S_y^h(x) = h_y(x) + h_{y+1}(x+1) - h_y(x+1) - h_{y+1}(x), \quad y = 0, 1, \quad x \geq 0, \quad (4.6)$$

$$M_y^h(x) = h_y(x+1) + h_{y+1}(x-1) - h_y(x) - h_{y+1}(x), \quad y = 0, 1, \quad x \geq 1, \quad (4.7)$$

$$R^h(x) = h_0(x) + h_2(x-2) - 2h_1(x-1), \quad x \geq 2. \quad (4.8)$$

The condition $h \in \mathcal{K}$ means exactly that all of these primary finite-difference quantities are nonnegative on their natural domains. In particular,

$$\textbf{(K1): } X_y^h(x) = h_y(x+1) - h_y(x) \geq 0, \quad y = 0, 1, 2, \quad x \geq 0, \quad (4.9)$$

$$\textbf{(K2): } C_y^h(x) = h_y(x+1) - 2h_y(x) + h_y(x-1) \geq 0, \quad y = 0, 1, 2, \quad x \geq 1, \quad (4.10)$$

$$\textbf{(K5): } Y_y^h(x) = h_{y+1}(x) - h_y(x) \geq 0, \quad y = 0, 1, \quad x \geq 0, \quad (4.11)$$

$$\textbf{(K6): } Z^h(x) = h_0(x) + h_2(x) - 2h_1(x) \geq 0, \quad x \geq 0, \quad (4.12)$$

$$\textbf{(K4): } S_y^h(x) = h_y(x) + h_{y+1}(x+1) - h_y(x+1) - h_{y+1}(x) \geq 0, \quad y = 0, 1, \quad x \geq 0, \quad (4.13)$$

$$\textbf{(K3): } M_y^h(x) = h_y(x+1) + h_{y+1}(x-1) - h_y(x) - h_{y+1}(x) \geq 0, \quad y = 0, 1, \quad x \geq 1, \quad (4.14)$$

$$\textbf{(K7): } R^h(x) = h_0(x) + h_2(x-2) - 2h_1(x-1) \geq 0, \quad x \geq 2. \quad (4.15)$$

To write reflected-boundary identities compactly, define the following auxiliary boundary extensions for every function h :

$$\widehat{C}_y^h(x) = \begin{cases} X_y^h(0), & x = 0, \\ C_y^h(x), & x \geq 1, \end{cases} \quad \widehat{M}_y^h(x) = \begin{cases} X_y^h(0), & x = 0, \\ M_y^h(x), & x \geq 1, \end{cases} \quad (4.16)$$

$$\widehat{S}_y^h(x) = \begin{cases} 0, & x = -1, \\ S_y^h(x), & x \geq 0, \end{cases} \quad \widehat{R}^h(x) = \begin{cases} X_0^h(0) + Z^h(0), & x = 1, \\ R^h(x), & x \geq 2. \end{cases} \quad (4.17)$$

These hatted quantities are not additional cone inequalities. They are only boundary extensions of the primary quantities. If $h \in \mathcal{K}$, then

$$\widehat{C}_y^h(x), \quad \widehat{M}_y^h(x), \quad \widehat{S}_y^h(x), \quad \widehat{R}^h(x) \quad (4.18)$$

are nonnegative wherever they are defined. The same definitions will be used for $h = f$ in the calculations below and for $h = g$ in the next induction step.

The event operator is

$$g_0(x) = \lambda f_0(x+1) + \mu_1 f_0((x-1)^+) + 2\mu_2 f_0(x), \quad (4.19)$$

$$g_1(x) = \lambda f_1(x+1) + \mu_1 f_1((x-1)^+) + \mu_2 f_0(x) + \mu_2 f_1(x), \quad (4.20)$$

$$g_2(x) = \lambda f_2(x+1) + \mu_1 f_2((x-1)^+) + 2\mu_2 f_1(x). \quad (4.21)$$

The only delicate point is the reflected fast-completion coordinate $(x-1)^+$. Each calculation below either separates or encodes the boundary contribution from this reflection. In particular, there is no hatted X -quantity: for the X -differences the reflected fast-completion contribution is genuinely zero at $x = 0$.

Monotonicity in x (K1). The desired inequalities are the primary inequalities for g ,

$$X_y^g(x) := g_y(x+1) - g_y(x) \geq 0, \quad y = 0, 1, 2, \quad x \geq 0. \quad (4.22)$$

For the reflected fast-completion term,

$$f_y(((x+1)-1)^+) - f_y((x-1)^+) = \begin{cases} 0, & x = 0, \\ X_y^f(x-1), & x \geq 1. \end{cases} \quad (4.23)$$

Thus the correct formulae are, for $x = 0$,

$$X_0^g(0) = \lambda X_0^f(1) + 2\mu_2 X_0^f(0), \quad (4.24)$$

$$X_1^g(0) = \lambda X_1^f(1) + \mu_2 X_0^f(0) + \mu_2 X_1^f(0), \quad (4.25)$$

$$X_2^g(0) = \lambda X_2^f(1) + 2\mu_2 X_1^f(0), \quad (4.26)$$

and, for $x \geq 1$,

$$X_0^g(x) = \lambda X_0^f(x+1) + \mu_1 X_0^f(x-1) + 2\mu_2 X_0^f(x), \quad (4.27)$$

$$X_1^g(x) = \lambda X_1^f(x+1) + \mu_1 X_1^f(x-1) + \mu_2 X_0^f(x) + \mu_2 X_1^f(x), \quad (4.28)$$

$$X_2^g(x) = \lambda X_2^f(x+1) + \mu_1 X_2^f(x-1) + 2\mu_2 X_1^f(x). \quad (4.29)$$

Equivalently, the fast-completion contribution is $\mu_1 \mathbf{1}_{\{x \geq 1\}} X_y^f(x-1)$. Every term on the right is nonnegative. Notice in particular that there is no $\mu_1 X_y^f(0)$ term at $x = 0$; the reflected fast-completion contribution is zero there.

Convexity in x (K2). The desired inequalities are

$$C_y^g(x) := g_y(x+1) - 2g_y(x) + g_y(x-1) \geq 0, \quad y = 0, 1, 2, \quad x \geq 1. \quad (4.30)$$

For the reflected fast-completion term,

$$f_y(x) - 2f_y((x-1)^+) + f_y((x-2)^+) = \begin{cases} X_y^f(0), & x = 1, \\ C_y^f(x-1), & x \geq 2. \end{cases} \quad (4.31)$$

Equivalently, the contribution is $\widehat{C}_y^f(x-1)$. Therefore, for $x \geq 1$,

$$C_0^g(x) = \lambda C_0^f(x+1) + \mu_1 \widehat{C}_0^f(x-1) + 2\mu_2 C_0^f(x), \quad (4.32)$$

$$C_1^g(x) = \lambda C_1^f(x+1) + \mu_1 \widehat{C}_1^f(x-1) + \mu_2 C_0^f(x) + \mu_2 C_1^f(x), \quad (4.33)$$

$$C_2^g(x) = \lambda C_2^f(x+1) + \mu_1 \widehat{C}_2^f(x-1) + 2\mu_2 C_1^f(x). \quad (4.34)$$

Again each right side is a nonnegative linear combination of primary or auxiliary cone quantities of f .

Monotonicity in y (K5). The desired inequalities are

$$Y_y^g(x) := g_{y+1}(x) - g_y(x) \geq 0, \quad y = 0, 1, \quad x \geq 0. \quad (4.35)$$

Here the reflected fast-completion term is evaluated at the same coordinate in both slow levels, so no exceptional boundary term is created. Direct expansion gives

$$Y_0^g(x) = \lambda Y_0^f(x+1) + \mu_1 Y_0^f((x-1)^+) + \mu_2 Y_0^f(x), \quad (4.36)$$

$$Y_1^g(x) = \lambda Y_1^f(x+1) + \mu_1 Y_1^f((x-1)^+) + \mu_2 Y_0^f(x). \quad (4.37)$$

Both right sides are nonnegative.

Convexity in y fixing x (K6). The desired inequality is

$$Z^g(x) := g_0(x) + g_2(x) - 2g_1(x) \geq 0, \quad x \geq 0. \quad (4.38)$$

The slow-completion terms cancel exactly, and the reflected fast-completion terms stay at the same coordinate. Hence

$$Z^g(x) = \lambda Z^f(x+1) + \mu_1 Z^f((x-1)^+), \quad (4.39)$$

which is nonnegative.

Increasing differences, S (K4). The desired inequalities are

$$S_y^g(x) := g_y(x) + g_{y+1}(x+1) - g_y(x+1) - g_{y+1}(x) \geq 0, \quad y = 0, 1, \quad x \geq 0. \quad (4.40)$$

For the reflected fast-completion term,

$$\begin{aligned} & f_y((x-1)^+) + f_{y+1}(x) - f_y(x) - f_{y+1}((x-1)^+) \\ &= \begin{cases} 0, & x = 0, \\ S_y^f(x-1), & x \geq 1. \end{cases} \end{aligned} \quad (4.41)$$

Thus this contribution is $\widehat{S}_y^f(x-1)$. The remaining terms give

$$S_0^g(x) = \lambda S_0^f(x+1) + \mu_1 \widehat{S}_0^f(x-1) + \mu_2 S_0^f(x), \quad (4.42)$$

$$S_1^g(x) = \lambda S_1^f(x+1) + \mu_1 \widehat{S}_1^f(x-1) + \mu_2 S_0^f(x). \quad (4.43)$$

These formulae include the boundary case $x = 0$ because $\widehat{S}_y^f(-1) = 0$.

Diagonal increasing differences, M (K3). The desired inequalities are

$$M_y^g(x) := g_y(x+1) + g_{y+1}(x-1) - g_y(x) - g_{y+1}(x) \geq 0, \quad y = 0, 1, \quad x \geq 1. \quad (4.44)$$

For the reflected fast-completion term,

$$\begin{aligned} & f_y(x) + f_{y+1}((x-2)^+) - f_y((x-1)^+) - f_{y+1}((x-1)^+) \\ &= \begin{cases} X_y^f(0), & x = 1, \\ M_y^f(x-1), & x \geq 2. \end{cases} \end{aligned} \quad (4.45)$$

Thus this contribution is $\widehat{M}_y^f(x-1)$. The slow-completion terms reduce to $C_0^f + M_0^f$ for $y = 0$ and to $C_1^f + M_0^f$ for $y = 1$. Therefore, for $x \geq 1$,

$$M_0^g(x) = \lambda M_0^f(x+1) + \mu_1 \widehat{M}_0^f(x-1) + \mu_2 \{C_0^f(x) + M_0^f(x)\}, \quad (4.46)$$

$$M_1^g(x) = \lambda M_1^f(x+1) + \mu_1 \widehat{M}_1^f(x-1) + \mu_2 \{C_1^f(x) + M_0^f(x)\}. \quad (4.47)$$

All quantities on the right are nonnegative.

Convexity in y fixing n , R (K7). The desired inequalities are

$$R^g(x) := g_0(x) + g_2(x-2) - 2g_1(x-1) \geq 0, \quad x \geq 2. \quad (4.48)$$

The reflected fast-completion contribution is

$$\begin{aligned} & f_0(x-1) + f_2((x-3)^+) - 2f_1((x-2)^+) \\ &= \begin{cases} X_0^f(0) + Z^f(0), & x = 2, \\ R^f(x-1), & x \geq 3. \end{cases} \end{aligned} \quad (4.49)$$

This is $\widehat{R}^f(x-1)$. The slow-completion terms give $2M_0^f(x-1)$. Hence, for $x \geq 2$,

$$R^g(x) = \lambda R^f(x+1) + \mu_1 \widehat{R}^f(x-1) + 2\mu_2 M_0^f(x-1). \quad (4.50)$$

The boundary case $x = 2$ is included because $\widehat{R}^f(1) = X_0^f(0) + Z^f(0) \geq 0$.

We have shown that every primary defining inequality of $\mathcal{K}$ is nonnegative for $g = \mathcal{P}f$. Hence $g \in \mathcal{K}$. For the next induction step, the auxiliary hatted quantities of g are then defined by the same generic rules stated at the beginning of the proof:

$$\widehat{C}_y^g(0) = X_y^g(0), \quad \widehat{M}_y^g(0) = X_y^g(0), \quad \widehat{S}_y^g(-1) = 0, \quad (4.51)$$

$$\widehat{R}^g(1) = X_0^g(0) + Z^g(0), \quad (4.52)$$

and away from the boundary they equal C_y^g , M_y^g , S_y^g , and R^g on their natural domains. Since $g \in \mathcal{K}$, all these hatted quantities for g are nonnegative. Thus the same notation and the same boundary-extension rules may be used at the next application of the event operator. If a suppressed uniformization self-transition is included, it simply adds a nonnegative multiple of the same primary inequality for f to the corresponding right side, and the conclusion is unchanged. $\square$

5 Boundary feasibility of the relaxed operator

The relaxed operator differs from the true nonpreemptive operator only when it would move a fast-server customer to a slow server. The relevant boundary comparisons are

$$V(1, 0) \leq V(0, 1), \quad (5.1)$$

$$V(1, 1) \leq V(0, 2). \quad (5.2)$$

If (5.1) holds, then the relaxed action $(1, 0) \mapsto (0, 1)$ is not better than keeping the customer at the fast server. If (5.2) holds, then neither $(1, 1) \mapsto (0, 2)$ nor the two-total-population move $(2, 0) \mapsto (0, 2)$ is needed; from $(2, 0)$ the feasible action $(2, 0) \mapsto (1, 1)$ is at least as good as $(2, 0) \mapsto (0, 2)$.

Lemma 4 (Boundary dominance). *Let $V_{m+1} = c + \rho \mathcal{P} \mathcal{Q} V_m$, with $V_m \in \mathcal{K}$. Then V_{m+1} satisfies (5.1) and (5.2).*

Proof. Let $W = \mathcal{Q} V_m$. By Lemma 2, $W \in \mathcal{K}$. Since W is a tail minimum, it satisfies the same-total tail-order inequalities

$$W(2, 0) \leq W(1, 1), \quad W(1, 0) \leq W(0, 1), \quad W(1, 1) \leq W(0, 2). \quad (5.3)$$

First compare $(1, 0)$ and $(0, 1)$. The immediate costs are both 1, so direct expansion gives

$$\begin{aligned} & V_{m+1}(0, 1) - V_{m+1}(1, 0) \\ &= \rho \left\{ \lambda [W(1, 1) - W(2, 0)] + (\mu_1 - \mu_2) [W(1, 0) - W(0, 0)] \right. \\ & \quad \left. + (\mu_1 + \mu_2) [W(0, 1) - W(1, 0)] \right\} \geq 0. \end{aligned} \quad (5.4)$$

The first and third bracketed terms are nonnegative by tail-order, and the second is nonnegative by monotonicity in x ; here $\mu_1 > \mu_2$ is used.

Next compare $(1, 1)$ and $(0, 2)$. The immediate costs are both 2, and expansion gives

$$\begin{aligned} & V_{m+1}(0, 2) - V_{m+1}(1, 1) \\ &= \rho \left\{ \lambda [W(1, 2) - W(2, 1)] + (\mu_1 - \mu_2) [W(1, 1) - W(0, 1)] \right. \\ & \quad \left. + \mu_1 [W(0, 2) - W(1, 1)] + \mu_2 [W(0, 1) - W(1, 0)] \right\} \geq 0. \end{aligned} \quad (5.5)$$

The first, third, and fourth bracketed terms are nonnegative by tail-order, and the second is nonnegative by monotonicity in x . Thus (5.1)–(5.2) hold for V_{m+1} . $\square$

Proposition 1 (The relaxed and true value recursions coincide). *Let $V_0 = 0$ and define $V_{m+1} = c + \rho \mathcal{P} \mathcal{Q} V_m$. For every m , $V_m \in \mathcal{K}$, V_m satisfies (5.1)–(5.2), and*

$$\mathcal{Q} V_m = \mathcal{Q}^{\text{true}} V_m. \quad (5.6)$$

Consequently the relaxed recursion and the original nonpreemptive recursion have the same finite-horizon value functions and admit feasible optimal selectors.

Proof. The zero function and the cost function lie in $\mathcal{K}$. Lemmas 2 and 3 imply by induction that $V_m \in \mathcal{K}$ for every m . Lemma 4 gives (5.1)–(5.2) after each step; they also hold for $V_0 = 0$.

The only infeasible relaxed arcs are the arcs that move a fast-server customer to a slow server. Inequality (5.1) rules out $(1, 0) \rightarrow (0, 1)$, because $V_m(1, 0) \leq V_m(0, 1)$. Inequality (5.2) rules out $(1, 1) \rightarrow (0, 2)$, because $V_m(1, 1) \leq V_m(0, 2)$. It also rules out the infeasible two-slow-server move $(2, 0) \rightarrow (0, 2)$, because the feasible one-slow-server move $(2, 0) \rightarrow (1, 1)$ has value no larger than $V_m(0, 2)$. Therefore every minimum in $\mathcal{Q} V_m$ is attained by a true feasible action, and $\mathcal{Q} V_m = \mathcal{Q}^{\text{true}} V_m$. $\square$

6 Threshold structure

Let V be any pointwise limit of value iterates, a discounted fixed point, or an average-cost bias limit obtained in Section 7. Since $\mathcal{K}$ is closed under pointwise limits, $V \in \mathcal{K}$. Define

$$A(n) = V(n, 0) - V(n - 1, 1), \quad n \geq 1, \quad (6.1)$$

$$B(n) = V(n - 1, 1) - V(n - 2, 2), \quad n \geq 2. \quad (6.2)$$

These are the same fixed-total differences as in Section 2. From (K3), $A(n)$ and $B(n)$ are nondecreasing in n , and from (K7),

$$A(n) \geq B(n). \quad (6.3)$$

Consider a state $(x, 1)$ with one idle slow server. The slow assignment is feasible only for $x \geq 2$. It is optimal to assign one queued customer to the idle slow server exactly when

$$V(x - 1, 2) \leq V(x, 1), \quad (6.4)$$

or, in total-population notation $n = x + 1$,

$$B(n) \geq 0. \quad (6.5)$$

Since B is nondecreasing, the set of feasible x for which this action is optimal is an upper set. At the infeasible boundary $x = 1$, inequality (5.2) says the relaxed move is dominated by keeping the customer at the fast server.

Now consider a state $(x, 0)$ with both slow servers idle. For $x \geq 3$, the three feasible alternatives have continuation values

$$H_0(n) = V(n, 0), \quad H_1(n) = V(n - 1, 1), \quad H_2(n) = V(n - 2, 2), \quad (6.6)$$

where $n = x$. Because $A \geq B$, exactly three ordered regimes are possible:

- if $A(n) < 0$, then $B(n) \leq A(n) < 0$, so $H_0(n) < H_1(n) < H_2(n)$ and no slow server is used;
- if $A(n) \geq 0$ and $B(n) < 0$, then $H_1(n) \leq H_0(n)$ and $H_1(n) < H_2(n)$, so exactly one slow server is used;
- if $B(n) \geq 0$, then $A(n) \geq B(n) \geq 0$, so $H_2(n) \leq H_1(n) \leq H_0(n)$ and both slow servers are used.

Since A and B are nondecreasing, these regimes are ordered by thresholds. For $x = 0, 1$, no slow assignment is feasible. For $x = 1$, the only relaxed slow-assignment candidate is the infeasible move $(1, 0) \mapsto (0, 1)$, and (5.1) gives $V(1, 0) \leq V(0, 1)$, so this candidate is dominated by the feasible choice of keeping the customer at the fast server. For $x = 2$, assigning two jobs to slow servers is infeasible, and (5.2) implies that the infeasible value $V(0, 2)$ is no smaller than the feasible one-slow value $V(1, 1)$; hence the feasible selector is still the restriction of the ordered relaxed selector. This proves the threshold assertion for finite-horizon and discounted value functions.

7 Discounted and average-cost optimality

Lemma 5 (Discounted and average-cost optimality). (a) *For every discount rate $\beta > 0$, the discounted value function belongs to $\mathcal{K}$, satisfies the boundary comparisons (5.1)–(5.2), and has an optimal stationary selector with the threshold structure of Section 6.*

(b) *If $\lambda < \mu_1 + 2\mu_2$, then the average-cost problem has an optimal stationary selector with the same threshold structure.*

Proof. Let

$$\nu = \lambda + \mu_1 + 2\mu_2. \quad (7.1)$$

For a continuous-time discount rate $\beta > 0$, let V_β denote the discounted value function in the original continuous-time model. The uniformized Bellman equation is

$$V_\beta = \frac{c}{\nu + \beta} + \frac{1}{\nu + \beta} \mathcal{P} \mathcal{Q}^{\text{true}} V_\beta. \quad (7.2)$$

Equivalently, if

$$U_\beta = (\nu + \beta) V_\beta, \quad (7.3)$$

then U_β is the unique fixed point of

$$\mathcal{T}_\beta f = c + \frac{1}{\nu + \beta} \mathcal{P} \mathcal{Q}^{\text{true}} f. \quad (7.4)$$

The positive rescaling in (7.3) does not change minimizers, does not change boundary comparisons, and preserves membership in $\mathcal{K}$ by the closure properties recorded in Section 2.

Put $\hat{P} = \nu^{-1} \mathcal{P}$. Then $\hat{P}$ is the one-step transition operator of the uniformized discrete-time chain, and

$$\rho_\beta = \frac{\nu}{\nu + \beta} \in (0, 1) \quad (7.5)$$

is its discount factor. Hence (7.4) can be written as

$$\mathcal{T}_\beta f = c + \rho_\beta \hat{P} \mathcal{Q}^{\text{true}} f. \quad (7.6)$$

This is the normalized discrete-time discounted operator to which we apply the standard countable-state theory. The scalar in the average-cost limit is unchanged by the rescaling, because

$$(1 - \rho_\beta) U_\beta(0, 0) = \beta V_\beta(0, 0). \quad (7.7)$$

We first prove the discounted statement. For $b > 0$, define

$$w_b(x, y) = b + x + y, \quad \|f\|_{w_b} = \sup_{(x, y)} \frac{|f(x, y)|}{w_b(x, y)}. \quad (7.8)$$

We prove the contraction estimate in detail. Let $\Delta = f - g$ and write $N(s) = x + y$ for $s = (x, y)$. From a post-decision state s , let $e \in E(s)$ be a uniformized event outcome and let

$$z = \eta_e(s) \quad (7.9)$$

be the pre-decision state immediately after that event. The true assignment operator is applied at this pre-decision state z , not at the original post-decision state s . Therefore

$$\begin{aligned} & \left| (\hat{P} \mathcal{Q}^{\text{true}} f)(s) - (\hat{P} \mathcal{Q}^{\text{true}} g)(s) \right| \\ &= \left| \sum_{e \in E(s)} p_e [(\mathcal{Q}^{\text{true}} f)(\eta_e(s)) - (\mathcal{Q}^{\text{true}} g)(\eta_e(s))] \right| \\ &\leq \sum_{e \in E(s)} p_e \left| (\mathcal{Q}^{\text{true}} f)(\eta_e(s)) - (\mathcal{Q}^{\text{true}} g)(\eta_e(s)) \right|. \end{aligned} \quad (7.10)$$

Fix one event outcome e and write $z = \eta_e(s)$. If a is an admissible assignment action at z , let $\psi(z, a)$ be the resulting post-decision state. Then

$$\begin{aligned} |\mathcal{Q}^{\text{true}} f(z) - \mathcal{Q}^{\text{true}} g(z)| &= \left| \min_{a \in A(z)} f(\psi(z, a)) - \min_{a \in A(z)} g(\psi(z, a)) \right| \\ &\leq \max_{a \in A(z)} |\Delta(\psi(z, a))|. \end{aligned} \quad (7.11)$$

The event can increase the total population by at most one, so

$$N(z) \leq N(s) + 1. \quad (7.12)$$

Assignment only moves queued customers to idle servers and therefore does not change total population. Hence, for every $a \in A(z)$,

$$N(\psi(z, a)) = N(z) \leq N(s) + 1. \quad (7.13)$$

It follows that

$$\begin{aligned} |\mathcal{Q}^{\text{true}} f(z) - \mathcal{Q}^{\text{true}} g(z)| &\leq \|\Delta\|_{w_b} \max_{a \in A(z)} w_b(\psi(z, a)) \\ &\leq \|\Delta\|_{w_b} [b + N(s) + 1] \\ &\leq \left(1 + \frac{1}{b}\right) \|\Delta\|_{w_b} w_b(s). \end{aligned} \quad (7.14)$$

Combining (7.10) with (7.14) and dividing by $w_b(s)$ gives, for all f, g ,

$$\|\rho_\beta \hat{P} \mathcal{Q}^{\text{true}} f - \rho_\beta \hat{P} \mathcal{Q}^{\text{true}} g\|_{w_b} \leq \rho_\beta \left(1 + \frac{1}{b}\right) \|f - g\|_{w_b}. \quad (7.15)$$

Choose $b > \nu/\beta$. Then $\rho_\beta(1 + 1/b) < 1$, and $\mathcal{T}_\beta$ is a contraction on the weighted supremum space. Thus the finite-horizon iterates of (7.4) converge to U_β .

Now start the relaxed iteration at $U_{\beta,0} = 0$ and define

$$U_{\beta,m+1} = c + \frac{1}{\nu + \beta} \mathcal{P} \mathcal{Q} U_{\beta,m}. \quad (7.16)$$

By Lemmas 2 and 3, by the closure properties recorded in Section 2, and because $c \in \mathcal{K}$, the iterates $U_{\beta,m}$ lie in $\mathcal{K}$. Lemma 4 applies with $\rho = (\nu + \beta)^{-1}$, so the boundary comparisons hold at every step. Proposition 1 then gives $\mathcal{Q} U_{\beta,m} = \mathcal{Q}^{\text{true}} U_{\beta,m}$ for every m , and hence the relaxed and true discounted iterates coincide. Passing to the weighted-norm limit and using the pointwise-closure property of $\mathcal{K}$, we get

$$U_\beta \in \mathcal{K}, \quad U_\beta(1, 0) \leq U_\beta(0, 1), \quad U_\beta(1, 1) \leq U_\beta(0, 2). \quad (7.17)$$

Therefore $V_\beta = (\nu + \beta)^{-1} U_\beta$ has the same properties and the same minimizing selectors. Section 6 proves the discounted threshold conclusion.

It remains to pass to average cost. We utilize the standard countable-state MDP theorem in Sennott [6]. We first explain why our post-decision MDP formulation fits into the standard MDP formulation, which is used in Sennott [6]. The state space is the post-decision state space $\mathcal{S}$. From a post-decision state s , let $E(s)$ be the finite set of possible uniformized event outcomes, let p_e be the probability of event e , and let $\eta_e(s)$ be the pre-decision state immediately after event e . If a_e is an

admissible assignment at $\eta_e(s)$, let $\psi(\eta_e(s), a_e)$ be the next post-decision state. A standard action at s is the event-contingent vector

$$a = (a_e)_{e \in E(s)}, \quad a_e \in A(\eta_e(s)). \quad (7.18)$$

This is an ordinary finite action set. The associated one-step transition kernel is

$$p(s' \mid s, a) = \sum_{e \in E(s)} p_e \mathbf{1}\{\psi(\eta_e(s), a_e) = s'\}. \quad (7.19)$$

Thus the standard discounted Bellman equation for this equivalent countable-state MDP is

$$U_\beta(s) = c(s) + \rho_\beta \min_{(a_e)} \sum_{e \in E(s)} p_e U_\beta(\psi(\eta_e(s), a_e)). \quad (7.20)$$

Because the components a_e are independent across event outcomes and the probabilities p_e are nonnegative, the minimization separates:

$$\begin{aligned} \min_{(a_e)} \sum_{e \in E(s)} p_e U_\beta(\psi(\eta_e(s), a_e)) \\ &= \sum_{e \in E(s)} p_e \min_{a_e \in A(\eta_e(s))} U_\beta(\psi(\eta_e(s), a_e)) \\ &= (\hat{P} \mathcal{Q}^{\text{true}} U_\beta)(s). \end{aligned} \quad (7.21)$$

Consequently the standard Bellman equation is the post-decision equation

$$U_\beta = c + \rho_\beta \hat{P} \mathcal{Q}^{\text{true}} U_\beta. \quad (7.22)$$

The relaxed operator $\mathcal{Q}$ is not used in applying Sennott; it is used only after the boundary-dominance argument has shown that $\mathcal{Q}U_\beta = \mathcal{Q}^{\text{true}}U_\beta$ for the discounted value functions. This event-contingent action formulation is the same standard device used in uniformized queueing-control models: one chooses, before the uniformized event is realized, the assignment rule to be used after each possible event outcome.

We use Sennott [6], Assumptions 1, 2, and 3*, and Theorems 1 and 2. The state space is countable, each action set in (7.18) is finite, and the one-stage cost c is nonnegative. With reference state

$$s_0 = (0, 0), \quad (7.23)$$

Sennott's hypotheses, in the notation of the present paper, are as follows.

- *Sennott's Assumption 1:* $U_\beta(s) < \infty$ for every state s and every discount factor $\rho_\beta \in (0, 1)$.
- *Sennott's Assumption 2:* there is a finite constant N_0 such that

$$U_\beta(s) - U_\beta(s_0) \geq -N_0, \quad s \in \mathcal{S}, \quad \rho_\beta \in (0, 1). \quad (7.24)$$

- *Sennott's Assumption 3*:* there are finite nonnegative constants M_s , $s \in \mathcal{S}$, such that

$$U_\beta(s) - U_\beta(s_0) \leq M_s, \quad s \in \mathcal{S}, \quad \rho_\beta \in (0, 1), \quad (7.25)$$

and, for every state s and every feasible event-contingent action a ,

$$\sum_{s'} p(s' \mid s, a) M_{s'} < \infty. \quad (7.26)$$

Under Assumptions 1, 2, and 3*, Sennott's Theorems 1 and 2 imply that along a sequence $\rho_{\beta_k} \uparrow 1$ there is a subsequence, again denoted by $\beta_k \downarrow 0$, and functions $\bar{h}$ and a scalar g such that

$$(1 - \rho_{\beta_k})U_{\beta_k}(s_0) \rightarrow g, \quad U_{\beta_k}(s) - U_{\beta_k}(s_0) \rightarrow \bar{h}(s) \quad (7.27)$$

for every state s , the average-cost optimality equation

$$g + \bar{h}(s) = c(s) + \hat{P}\mathcal{Q}^{\text{true}}\bar{h}(s) \quad (7.28)$$

holds, and every stationary selector attaining the minimum in (7.28) is expected average-cost optimal.

We now verify Sennott's assumptions. Assumption 1 follows from the weighted-norm contraction above. Moreover, (7.17) and monotonicity in $\mathcal{K}$ imply that $U_{\beta}(s_0)$ is the smallest discounted value. Hence Assumption 2 holds with

$$N_0 = 0. \quad (7.29)$$

To verify Assumption 3* directly, assume

$$\lambda < \mu_1 + 2\mu_2. \quad (7.30)$$

Let $N(x, y) = x + y$, and consider the stationary full-service policy F that uses the fast server whenever possible and assigns queued customers to all idle slow servers whenever possible. Let $M = \mu_1 + 2\mu_2$ and $\varepsilon = M - \lambda > 0$. Under this policy, all three servers are busy whenever $N \geq 3$. For the uniformized discrete-time total-population chain, the drift of N^2 satisfies, for $N \geq 3$,

$$\begin{aligned} E^F[N_{m+1}^2 - N_m^2 \mid N_m = N] &= \frac{\lambda}{\nu}[(N+1)^2 - N^2] + \frac{M}{\nu}[(N-1)^2 - N^2] \\ &= \frac{-2\varepsilon N + \lambda + M}{\nu}. \end{aligned} \quad (7.31)$$

Consequently there are constants $b_0, b_1 < \infty$ such that, for all states,

$$E^F[N_{m+1}^2 - N_m^2 \mid (X_m, Y_m) = (x, y)] \leq b_0 - b_1 N(x, y). \quad (7.32)$$

Let τ_0 be the first hitting time of $s_0 = (0, 0)$ under the full-service policy. We now show that the Foster-Lyapunov inequality (7.32) implies that the expected accumulated cost up to τ_0 is finite. Choose R so large that

$$b_0 - b_1 n \leq -\frac{b_1}{2}n, \quad n > R, \quad (7.33)$$

and let

$$D_R := \{s \in \mathcal{S} : N(s) \leq R\}, \quad \tau_R := \inf\{m \geq 0 : (X_m, Y_m) \in D_R\}. \quad (7.34)$$

Applying optional sampling to the nonnegative process N_m^2 stopped at $\tau_R \wedge t$ and using (7.33) gives

$$E_s^F \left[\sum_{m=0}^{\tau_R \wedge t - 1} N(X_m, Y_m) \right] \leq \frac{2}{b_1} N(s)^2, \quad s \notin D_R. \quad (7.35)$$

Letting $t \rightarrow \infty$ and using monotone convergence yields

$$E_s^F \left[\sum_{m=0}^{\tau_R - 1} N(X_m, Y_m) \right] < \infty, \quad s \in \mathcal{S}. \quad (7.36)$$

It remains to pass from the finite set D_R to the particular state s_0 . Since D_R is finite and the full-service chain is irreducible on finite sublevel sets of N , from each state in D_R there is a path

of positive probability to s_0 that stays in a finite sublevel set. Hence there exist numbers $L_R < \infty$ and $p_R > 0$ such that, whenever the chain starts in D_R , it hits s_0 within the next L_R steps with probability at least p_R , along a path on which the total population remains bounded by a finite constant depending only on R . If this attempt fails, the state after the attempt lies in a finite sublevel set of N ; by (7.36), the expected accumulated value of N until the next return to D_R is finite, uniformly over that finite set. The strong Markov property at successive returns to D_R therefore gives a geometric-trials bound, and hence

$$\max_{r \in D_R} E_r^F \left[\sum_{m=0}^{\tau_0-1} N(X_m, Y_m) \right] < \infty. \quad (7.37)$$

Combining (7.36) with (7.37) gives, for every initial state s ,

$$C_s := E_s^F \left[\sum_{m=0}^{\tau_0-1} N(X_m, Y_m) \right] < \infty, \quad s \in \mathcal{S}. \quad (7.38)$$

For each fixed initial state s , the number C_s is a deterministic finite constant depending only on s and on the model parameters; it does not depend on the discount factor ρ_β .

Now compare the optimal discounted cost from s with the policy that follows F until τ_0 and then switches to an optimal discounted policy from s_0 . For every $\rho_\beta \in (0, 1)$,

$$\begin{aligned} U_\beta(s) &\leq E_s^F \left[\sum_{m=0}^{\tau_0-1} \rho_\beta^m N(X_m, Y_m) + \rho_\beta^{\tau_0} U_\beta(s_0) \right] \\ &\leq E_s^F \left[\sum_{m=0}^{\tau_0-1} N(X_m, Y_m) \right] + U_\beta(s_0) \\ &= C_s + U_\beta(s_0). \end{aligned} \quad (7.39)$$

Thus (7.25) in Assumption 3* holds with

$$M_s = C_s, \quad s \in \mathcal{S}. \quad (7.40)$$

It remains only to verify the additional summability requirement (7.26). Fix a state s and a feasible event-contingent action a . The uniformized event set $E(s)$ is finite, and each component of a maps its corresponding event outcome to a single post-decision state. Hence the set

$$\{s' : p(s' | s, a) > 0\} \quad (7.41)$$

is finite. Since $M_{s'} = C_{s'} < \infty$ for every individual state s' , we have

$$\sum_{s'} p(s' | s, a) M_{s'} = \sum_{s' : p(s' | s, a) > 0} p(s' | s, a) C_{s'} < \infty. \quad (7.42)$$

Therefore Sennott's Assumptions 1, 2, and 3* are verified.

By Sennott's Theorems 1 and 2, there are g and $\bar{h}$ satisfying (7.27)–(7.28), and any stationary selector attaining the minimum in (7.28) is average-cost optimal. Since $U_\beta \in \mathcal{K}$ for every β , and $\mathcal{K}$ is closed under subtracting constants and pointwise limits, the limit $\bar{h}$ belongs to $\mathcal{K}$ and satisfies the boundary comparisons

$$\bar{h}(1, 0) \leq \bar{h}(0, 1), \quad \bar{h}(1, 1) \leq \bar{h}(0, 2). \quad (7.43)$$

Define the continuous-time bias

$$h = \frac{\bar{h}}{\nu}. \quad (7.44)$$

Because $\mathcal{K}$ is a cone, $h \in \mathcal{K}$ and the boundary comparisons remain true. Multiplying (7.28) by one and substituting $\bar{h} = \nu h$ gives the continuous-time average-cost optimality equation

$$g + \nu h(s) = c(s) + \mathcal{P}Q^{\text{true}}h(s), \quad s = (x, y). \quad (7.45)$$

The scalar g is the optimal continuous-time average cost by (7.7) and (7.27). Since h satisfies the boundary comparisons, the relaxed and true minimizing actions agree at the boundary. Section 6 applied to h gives an average-cost optimal stationary selector with the same threshold structure. This completes the proof of Theorem 1. $\square$

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